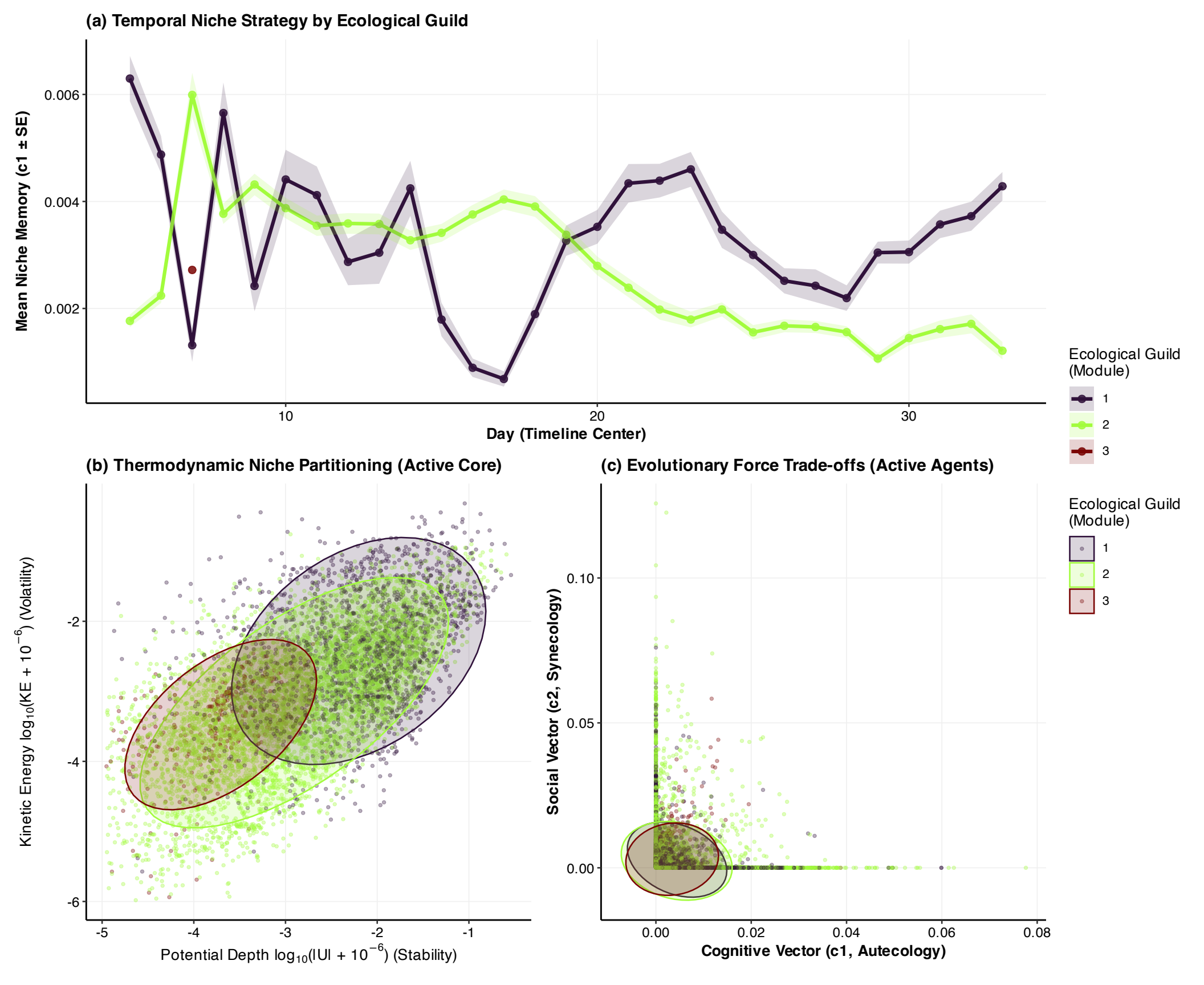

(a) Temporal Niche Strategy by Ecological Guild
Mean Niche Memory (c1 ± SE)
0.006
0.004
0.002
10
20
30
Day (Timeline Center)
Ecological Guild (Module)
1
2
3
(b) Thermodynamic Niche Partitioning (Active Core)
Kinetic Energy log10(KE + 10−6) (Volatility)
-2
-4
-6
-5
-4
-3
-2
-1
Potential Depth log10(|U| + 10−6) (Stability)
(c) Evolutionary Force Trade-offs (Active Agents)
Social Vector (c2, Synecology)
0.10
0.05
0.00
0.00
0.02
0.04
0.06
0.08
Cognitive Vector (c1, Autecology)
Ecological Guild (Module)
1
2
3